SAILY (NORD SECURITY) | SENIOR DATA ANALYST CASE STUDY

Unlocking Android Conversion Parity, Stopping Hardware CAC Bleed & Capturing Multi-Country Demand

End-to-End PACE Framework Synthesis (Plan -> Analyze -> Construct -> Execute)
Observation Window: Aug 1 – Aug 27, 2024 | 241,548 Deduplicated Sessions | 25,133 Paid Orders | $358,625.85 Gross Billings

- Part I — Core Executive Deck (Slides 1–7): Direct Answers to the 5 Case Study Questions & P0–P2 Action Roadmap
- Part II — Technical Deep-Dive Appendix (Slides 8–12): H1–H8 Registry, Simpson's Paradox, Behavioral Topology, Quota Matrix & Telemetry v2

Candidate: Mantas Normantas | Role Applied: Data Analyst (Saily) | Data Engine: Google Cloud BigQuery

True Deduplicated Session CR is 10.12% (+0.97 pp vs Naive); Closing Android Gaps Unlocks +$297k to +$1.40M/yr

1. TRUE SESSION CONVERSION
10.12% Strict CR
24,456 buyers / 241,548 sessions
(vs 9.15% naive row CR; 10.40% order ratio)

2. iOS vs ANDROID COMPAT (H1)
+2.84 pp iOS Lead
iOS 12.48% vs And. Compat 9.65%
(Z = 20.28, p < 0.0001; S1->S2 & S2->S3)

3. AUG 27 MOROCCO SURGE (H3)
11.36% Ex-MA CR
5,134 MA iOS sessions (0.10% CR)
diluted global iOS CR to 5.64%

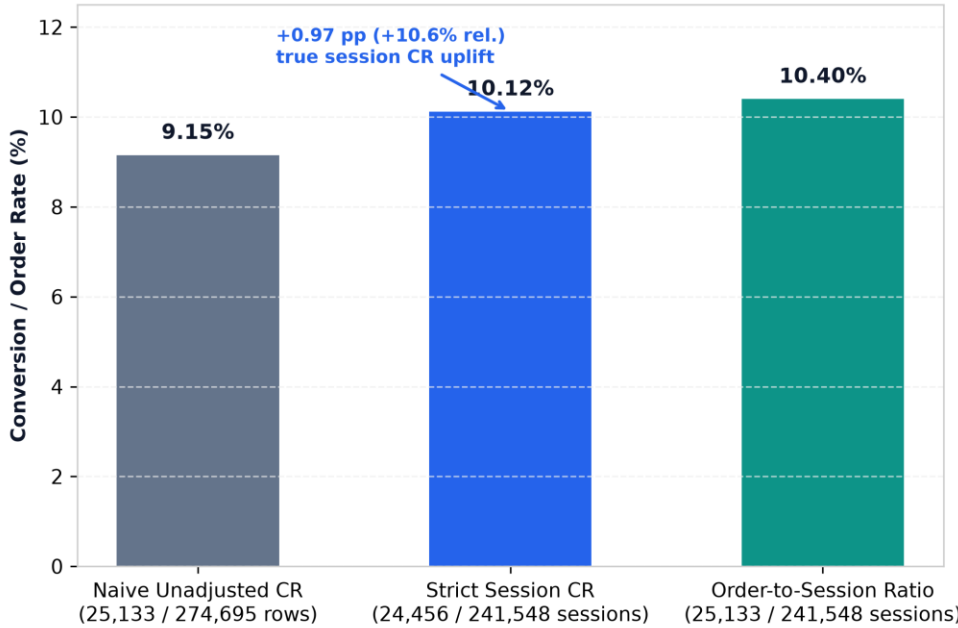
4. INCOMPATIBLE ANDROID (H2/H4)
+$35.8k / -$39.4k
+$35,752 27d hardware ceiling (~$483k/yr)
-$39,371 27d Paid CAC bleed (-$532k/yr)

Case Study Question	Verified Empirical Finding (BigQuery Reconciled)	Primary Root Cause	Priority Action & Annualized P&L Impact
1. Overall CR & Data Hygiene	Strict Session CR = 10.12% (24,456 / 241,548) Order Ratio = 10.40% (25,133 orders; $358,625.85) Naive Row CR = 9.15% (25,133 / 274,695 raw rows)	30,836 duplicate session IDs generate 33,147 surplus rows; 677 multi-plan orders occur in same session.	Deploy mart_saily_sessions_deduped in dbt & split fct_sessions / fct_orders (Telemetry v2).
2. Cohort Funnel Comparison (H1)	iOS = 12.48% CR (16,088 / 128,872) Android Compat = 9.65% CR (8,258 / 85,603) Android Incompat = 0.41% CR (110 / 27,073)	Android Compat wins S0->S1 (52.18% vs 48.52%), but suffers 43.42% drop at S1->S2 & 67.32% at S2->S3.	P1-1: Redesign Android S1->S2 config UX, surface $/GB badges & promo parity (+$39k to +$134k/yr).
3. Incompatible Drop-Off (H2)	70.78% (19,162) bounce at S0; 7,889 reach S1; 4,735 reach S2 (97.68% drop); 110 sessions ($1,238.19) pay.	Missing pre-checkout EuiccManager hardware gate allows 4,735 incompatible phones into checkout.	P0-2: Hard-block S2 on incompatible Android & show S0 modal (eliminates ~1,500 defective orders/yr).
4. Anomalies & Cyclicality (H3)	Aug 27 iOS CR = 5.64% observed vs 11.36% Ex-Morocco. Aug 16 Peak = 1,130 orders ($16,240.64; 14.83% iOS CR).	Aug 27: 5,134 Moroccan iOS sessions (97.8% domestic MA search). Fri (1.083x) & Sat (1.074x) weekend peaks.	P1-3 & P2-2: Paid UA geo-anomaly circuit-breaker + shift ad pacing +15% to Thu PM–Sat AM (+$61k–$101k/yr).
5. Opportunity vs CAC Bleed (H4)	Upper-Bound Revenue Ceiling: +$35,751.65 / 27d. Real-World Paid CAC Bleed: -$39,371.31 / 27d (@ $1.50 CAC).	Saily cannot remotely upgrade user hardware; acquiring 27,073 incompatible sessions burns paid UA budget.	P0-1: Enable Play Console euicc catalog filter + Paid UA OEM exclusion (saves +$258k to +$1.26M/yr).

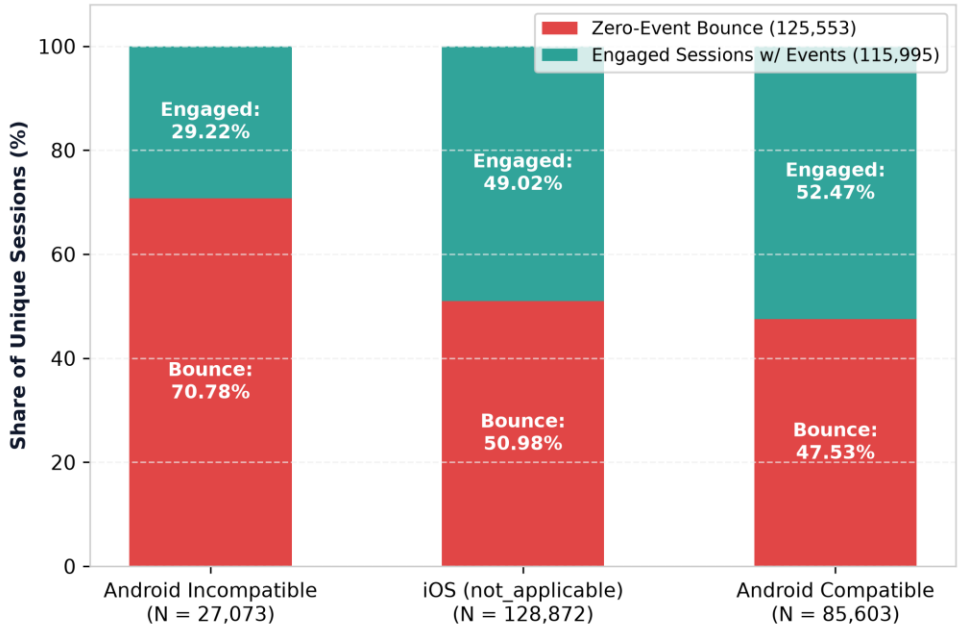
Collapsing 33,147 Duplicate Session Rows Lifts True Session CR +0.97 pp to 10.12% While Preserving All 25,133 Orders

Module 1 (DAMA Audit): Primary Key Deduplication & Zero-Event Bounce Structure

A. Impact of Session Deduplication on Top-Line CR (30,836 Duplicate Session IDs / 33,147 Surplus Rows)



B. Deduplicated Session Bounce Rate by Cohort (Incompatible Android Bounces +23.25 pp Higher)



PRIMARY KEY DEDUPLICATION

241,548 Sessions

274,695 raw rows contain 30,836 duplicate IDs (33,147 surplus rows; +20.9% JOIN inflation risk)

MULTI-ORDER PRESERVATION

25,133 Paid Orders

$358,625.85 billings ($14.27 AOV) across 24,456 buyers (677 multi-plan orders; $14.66 ARPPU)

ZERO-EVENT BOUNCE GAP

70.78% Incompat Bounce

125,553 bounced sessions (51.98% global): 70.78% And. Incompat vs 47.53% And. Compat

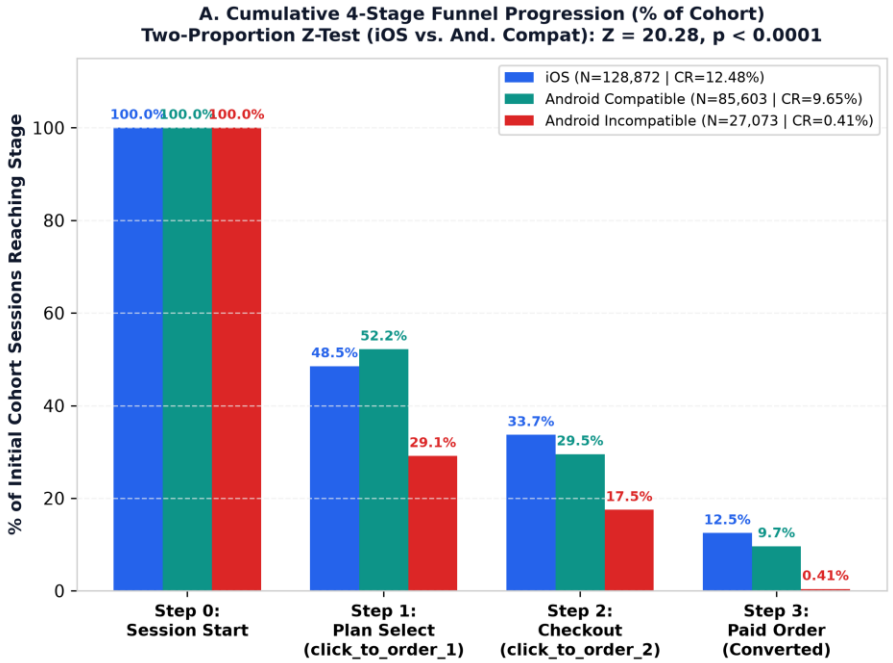
TAG SYNC & CLIENT LOOPS

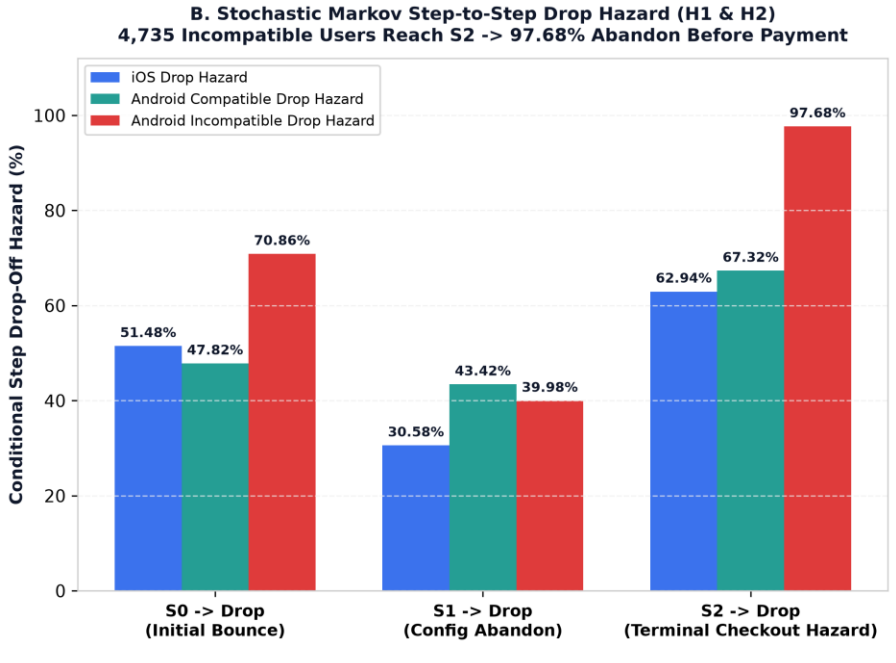
99.88% Tag Sync

24,427 / 24,456 buyers logged events (29 dark); 250 sessions fired >15 identical taps (max 167)

Android Compat Wins S0->S1 (52.2% vs 48.5%), Drops at S1->S2 (43.4%) & S2->S3 (67.3%); 97.7% Incompat Wall

Module 2: Multi-Cohort Conversion Funnel & Stochastic Markov Drop Hazard Dynamics

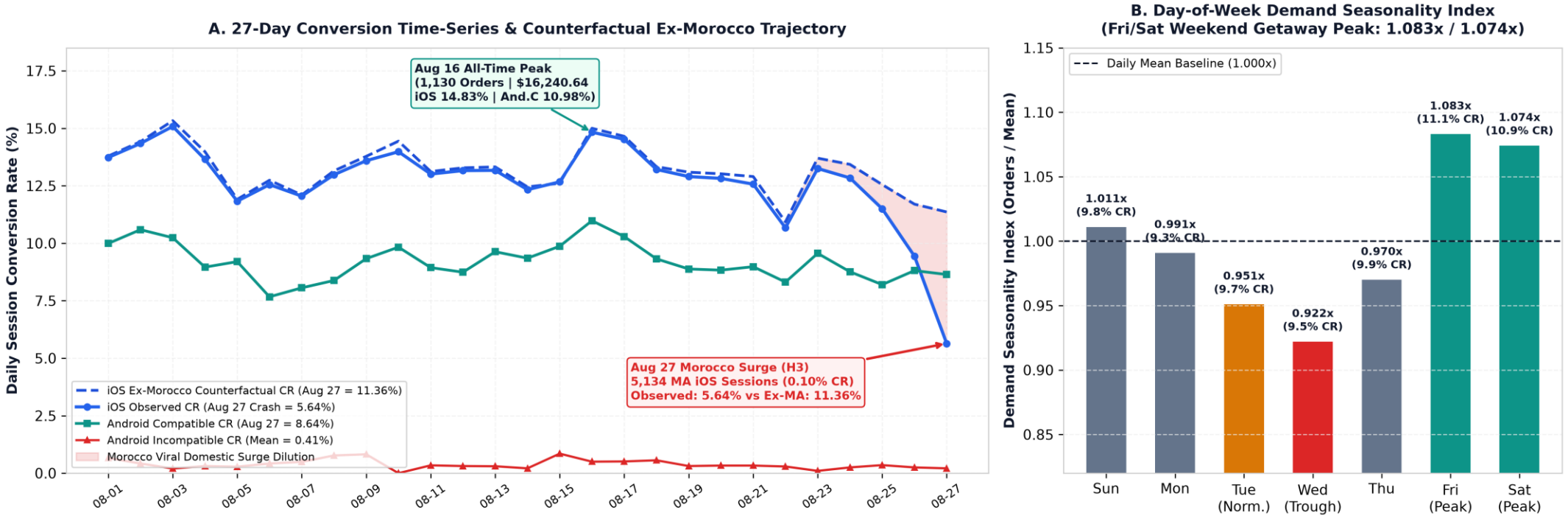




Device & Compatibility Cohort	S0: Unique Sessions	S1: Plan Select (click_to_order_1)	S2: Checkout (click_to_order_2)	S3: Buyers (Paid Orders)	S0->S1 Reach	S1->S2 Drop Hazard	S2->S3 Drop Hazard	Strict CR (Order Ratio)
iOS (not_applicable)	128,872	62,532	43,411	16,088 (16,527)	48.52%	30.58%	62.94%	12.48% (12.82%)
Android Compatible (true)	85,603	44,666	25,273	8,258 (8,495)	52.18%	43.42%	67.32%	9.65% (9.92%)
Android Incompatible (false)	27,073	7,889	4,735	110 (111)	29.14%	39.98%	97.68%	0.41% (0.41%)
Global Total (Deduplicated)	**241,548**	**115,087**	**73,419**	**24,456 (25,133)**	**47.65%**	**36.21%**	**66.69%**	**10.12% (10.40%)**

Aug 27 iOS Crash (5.64%) Was Driven by 5,134 Moroccan Domestic Sessions (0.10% CR); Ex-Morocco iOS CR Was 11.36%

Module 3: Macro Anomaly Deconstruction (Aug 27 Morocco & Aug 16 Peak) & Weekly Cyclicality



AUG 27 MOROCCO SURGE (H3)

5,134 MA iOS Sessions

50.79% of Aug 27 iOS traffic (10,109 total); converted at 0.10% (5 buyers) on $3.99 tourist rate

DOMESTIC MA SEARCH INTENT

97.8% Local MA Search

Across all 5,787 MA sessions on Aug 27, 5,019 of 5,134 logged event pairs searched destination = MA

COUNTERFACTUAL EX-MA CR

11.36% Ex-MA iOS CR

Excluding Morocco restores Aug 27 iOS CR from 5.64% to 11.36% (565 / 4,975) — zero app bug

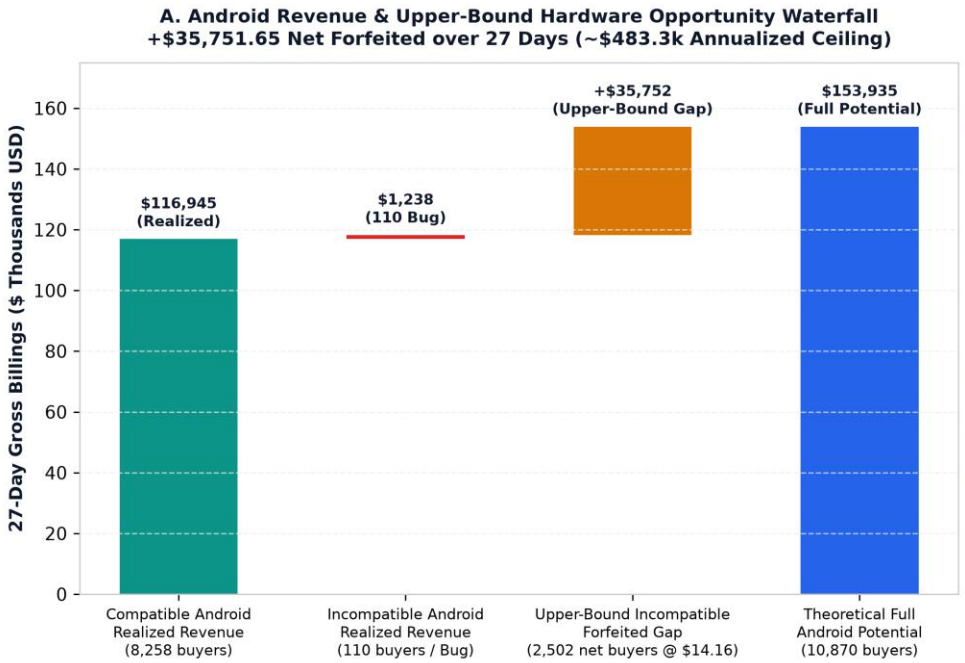
AUG 16 PEAK & CYCLICALITY

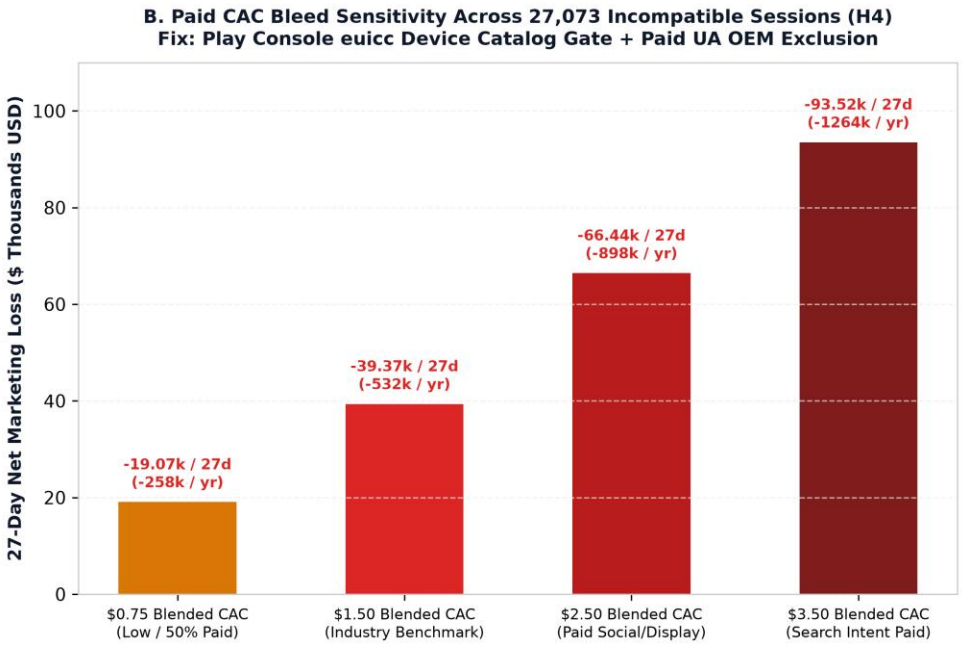
1.083x Fri / 1.074x Sat

Aug 16 peak: 1,130 orders ($16,240.64; 14.83% iOS CR).
Fri (11.10% CR) & Sat (10.89% CR) vs Wed (0.922x)

Upper-Bound Hardware Parity Equals +$35,752/27d (~$483k/yr), But Real-World Paid CAC Bleed Destroys -$532k/yr

Module 4: Addressable Android Hardware Opportunity Sizing & Paid CAC Bleed Sensitivity (H4)





Valuation / Sensitivity Scenario	Session / Buyer Volume (27 Days)	Gross Revenue / Spend Formula	Net 27-Day P&L Impact	Annualized Impact (365/27)
Upper-Bound Hardware Revenue Ceiling (H2)	27,073 sessions -> 2,612 buyers (2,502 net)	2,612 x $14.1615 ARPPU ($36,989.84) - $1,238.19 actual	+$35,751.65 forfeited	+$483,310 / yr ceiling
Paid CAC Bleed — Low ($0.75 Blended CAC)	27,073 incompat sessions (110 bug buyers)	$20,304.75 ad spend vs $1,238.19 defective revenue	-$19,066.56 net loss	-$257,752 / yr bleed
Paid CAC Bleed — Base ($1.50 Blended CAC)	27,073 incompat sessions (110 bug buyers)	$40,609.50 ad spend vs $1,238.19 defective revenue	-$39,371.31 net loss	-$532,240.64 / yr bleed
Paid CAC Bleed — Mid/High ($2.50–$3.50 CAC)	27,073 incompat sessions (110 bug buyers)	$67,682.50–$94,755.50 spend vs $1,238.19 revenue	-$66.4k to -$93.5k loss	-$898k to -$1.26M / yr

Executing P0–P2 Initiatives Delivers +$258k–$1.26M/yr in CAC Savings & +$262k–$486k/yr in Incremental Gross Billings

P0-1: STOP PAID CAC BLEED
+$532k/yr Base Saved
Play Console euicc catalog gate + Paid UA non-eSIM OEM exclusion (+$258k–$1.26M/yr)

P0-2: BLOCK DEFECTIVE ORDERS
~1,500 Orders/yr Blocked
In-app EuiccManager S0 modal & S2 hard-block eliminates 110 bug buyers ($1,238 / 27d)

P1: ANDROID UX, PROMO & RETRY
+$128k–$262k/yr Lift
P1-1 Android S1->S2 UX & promo parity (+$39k–$134k)
+ P1-2 Checkout retry fallback (+$89k–$128k)

P2: BUNDLES & PEAK PACING
+$134k–$224k/yr Lift
P2-1 EU<->AT regional smart card (+$73k–$123k)
+ P2-2 Thu–Sat pacing & 10GB anchor (+$61k–$101k)

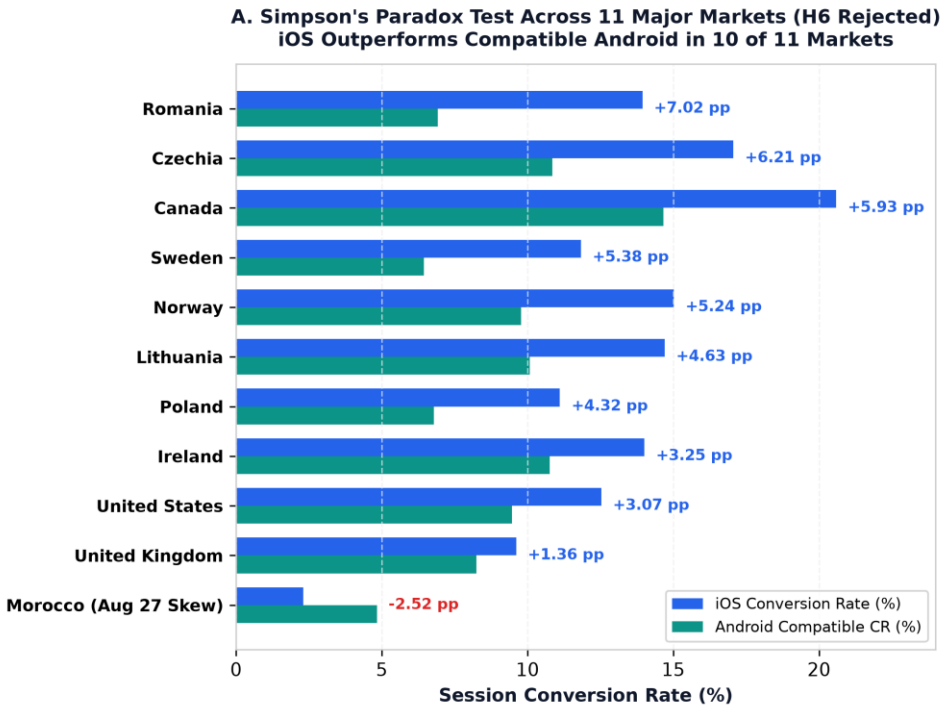
Priority	Strategic Initiative	Target Gap / Root Cause	Concrete Engineering / Growth Action	Annualized Impact (365/27)	Owner & Timeline
P0-1	Pre-Install Paid UA & Play Store Gate	27,073 incompat sessions (0.41% CR, 70.78% bounce)	Enable android.hardware.telephony.euicc in Play Console + exclude non-eSIM OEMs in Paid UA.	+$258k to +$1.26M/yr saved (+$532k base)	Perf. Mktg & Android Eng (Wk 1)
P0-2	In-App EuiccManager Gate (S0 + S2)	4,735 incompat reach S2; 110 sessions ($1,238.19) pay	Check EuiccManager.isEnabled() at S0; hard-block S2 checkout unless buying for companion device.	Eliminates ~1,500 defective orders & CS tickets/yr	Android Mobile Eng (Wk 1–2)
P1-1	Android S1->S2 UX & Promo Parity	43.42% Android S1->S2 drop; -4.43 pp promo gap	Sticky S1->S2 bottom sheet, $/GB savings badges (-39% to -56%), and Android promo deeplink parity.	+$39k to +$134k/yr net incremental billings	Product Growth & UX (Sprint 2–3)
P1-2	Checkout Retry Loop Fallback Sheet	iOS 3+ click_to_order_2 retries plunge -52.2% to 17.92%	Auto-trigger 1-Tap Payment Fallback Sheet (Apple/Google Pay <-> Card/PayPal) on 2nd S2 tap.	+$89k to +$128k/yr recovered checkouts	Payments Eng (Sprint 2–3)
P1-3	Paid UA Geo-Anomaly Circuit-Breaker	Aug 27 Morocco viral spike (5,134 iOS sessions, 0.10% CR)	Auto-pause ad sets when any country spikes >3x 7d volume at <1% CR; report Ex-Anomaly CR in BI.	Protects Paid UA ROAS & KPI integrity	Mktg Analytics & BI (Sprint 2)
P2-1	Regional vs Single-Country Smart Card	3+ countries = -72.3% CR drop; EU<->AT #1 pair (693)	Show 'Visiting 2+ EU Countries? Compare EU ($16.65) vs Single-Country ($9.79)' in 2–3 event window.	+$73k to +$123k/yr rescued multi-country orders	Search & Discovery (Sprint 4)
P2-2	Thu–Sat Peak Pacing & 10 GB Anchor	Fri (1.083x) / Sat (1.074x) peak; Transatlantic $17.28 AOV	Shift Paid UA budget +15% to Thu PM–Sat AM; default-select 10 GB anchor on US/CA -> EU corridors.	+$61k to +$101k/yr AOV & ROAS expansion	Growth Mktg & Pricing (Sprint 3)
ENG-v2	Telemetry v2 & dbt Warehouse Spec	33,147 dup rows, 125.5k 0-event bounces, no timestamps	Split fct_sessions / fct_orders; add user_id, event_timestamp, app_open, promo_code_id & 500ms debounce.	Eliminates -0.97 pp CR distortion & 52% blind spot	Data Eng & Mobile SDK (Q1)

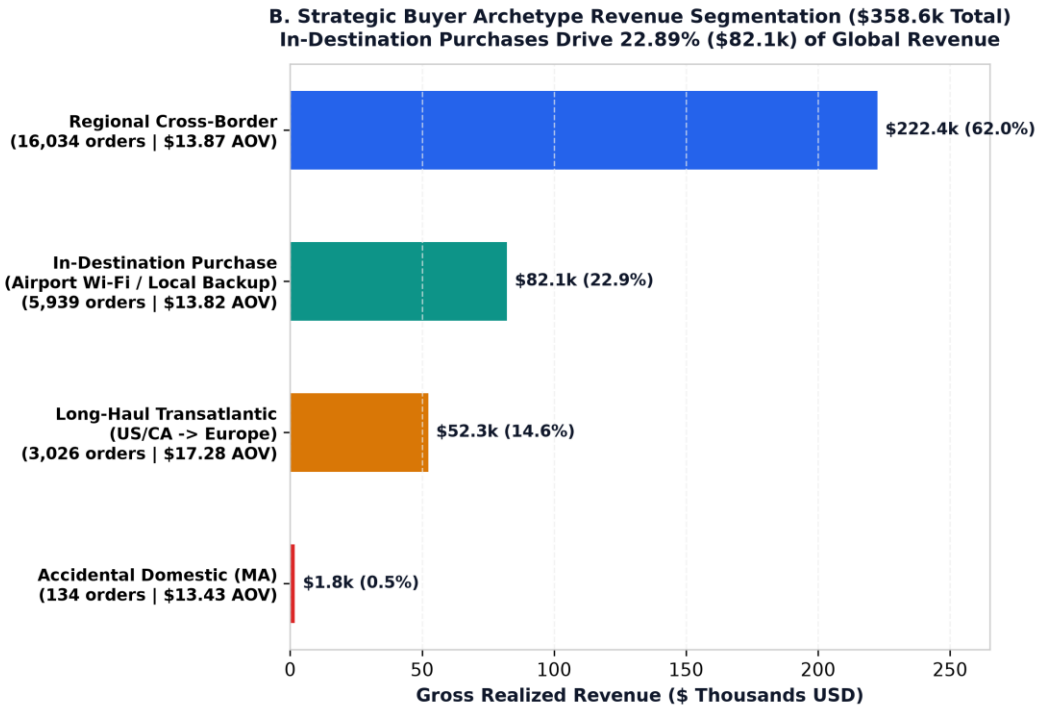
6 of 8 Hypotheses Empirically Confirmed; H5 (Quota Down-Selling) & H6 (Simpson's Paradox) Formally Rejected

ID	Hypothesis Statement	Statistical / SQL Test Method	Verified Empirical Evidence	Verdict
H1	iOS converts significantly higher than Compatible Android	Two-Proportion Z-Test (Deduplicated Sessions)	12.48% (16,088/128,872) vs 9.65% (8,258/85,603); Diff = +2.84 pp (Z = 20.28, p < 0.0001)	CONFIRMED
H2	Incompatible Android users reach payment before dropping	4-Stage Markov Transition Matrix (sql/02)	7,889 reach S1; 4,735 reach S2 (97.68% drop hazard); 110 sessions ($1,238.19) pay	CONFIRMED
H3	Aug 27 iOS crash was an external geo-composition shock	Country Decomposition & Counterfactual CR	5,134 Moroccan iOS sessions (0.10% CR; 97.8% domestic MA search); Ex-MA CR = 11.36%	CONFIRMED
H4	Paid UA on incompatible Android destroys net marketing margin	Blended CAC Sensitivity Matrix ($0.75–$3.50)	27,073 incompat sessions burn -$19.07k to -$93.52k net over 27d (-$39.37k / -$532k/yr base)	CONFIRMED
H5	Checkout drop-off is driven by mid-session quota down-selling	Timestamp-Free Quota Exploration Matrix (N=24,426)	92.16% (22,512) view 1 quota; 4.96% (1,211) view multiple & buy largest (97.12% diagonal)	REJECTED
H6	Global iOS CR lead is a Simpson's Paradox country mix artifact	Within-Country OS Stratification (Top 11 Markets)	iOS beats Compatible Android in 10 of 11 top markets (US +3.07 pp, CA +5.93 pp, RO +7.02 pp)	REJECTED
H7	iOS benefits from higher promotional discount penetration	Modal SKU Anchor Price Decomposition (N=24,782)	25.50% of iOS orders discounted ($1.29/order) vs 21.07% on Android ($1.14/order; +4.43 pp gap)	CONFIRMED
H8	Multi-country browsing & search loops trigger fatigue decay	Event Depth & Deterministic Cross-Shop Matrix	92.57% convert in 2–3 events (30.37% CR); 3+ countries = -72.3% CR drop; EU<->AT #1 pair (693)	CONFIRMED

iOS Beats Compatible Android in 10 of 11 Top Markets; In-Destination Purchases Drive 22.89% ($82.1k) of Billings

Module 3: Geographic Confounding Check (H6) & Global Buyer Archetype Economics





H6 REJECTED: 10 OF 11 MARKETS

iOS Leads Across Geos

US (+3.07 pp), UK (+1.36 pp), CA (+5.93 pp), PL (+4.32 pp), RO (+7.02 pp), CZ (+6.21 pp)

1. REGIONAL CROSS-BORDER

62.03% of Revenue

16,034 orders ($222,439.80, $13.87 AOV)
Core intra-European & regional travel

2. IN-DESTINATION PURCHASE

22.89% ($82,100.94)

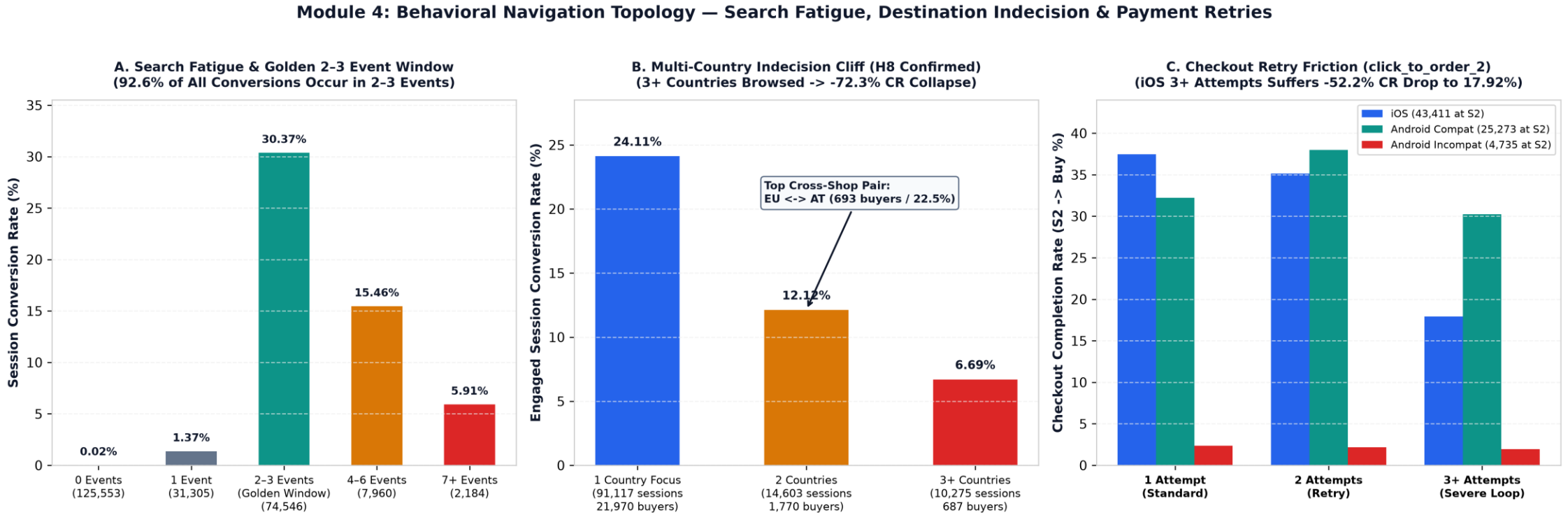
5,939 orders ($13.82 AOV; US->US $27.9k, GB->GB $13k)
Airport Wi-Fi arrival & local backup SIMs

3. LONG-HAUL TRANSATLANTIC

$17.28 AOV (+25.0%)

3,026 orders ($52,285.60 / 14.58% share)
Led by US->EU ($19.29 AOV) & US->GB ($16.43)

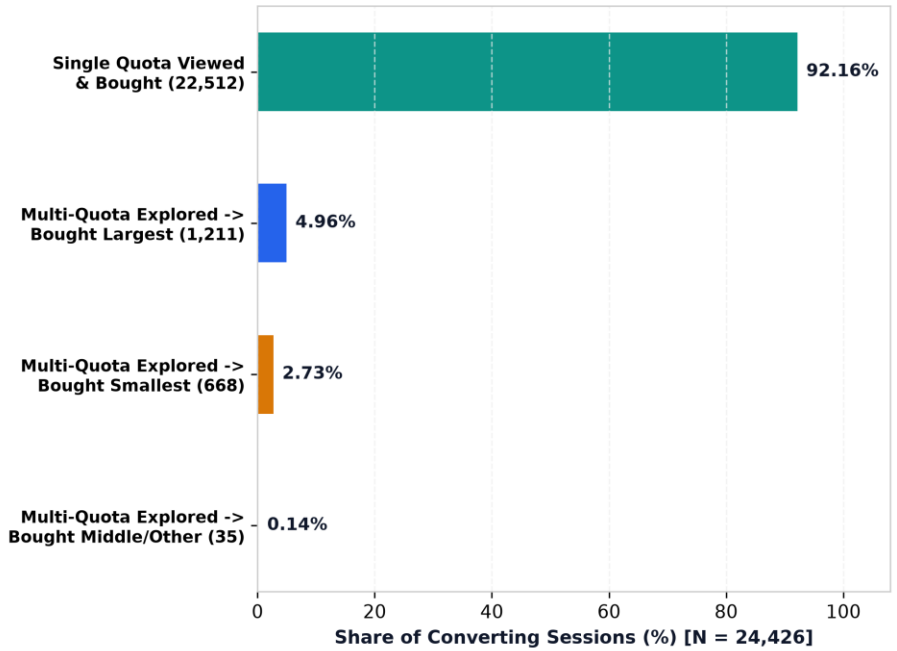
92.57% of Buyers Convert in 2–3 Events (30.37% CR); 3,078 Buyers Cross-Shopped Destinations (EU <-> AT #1 at 22.51%)

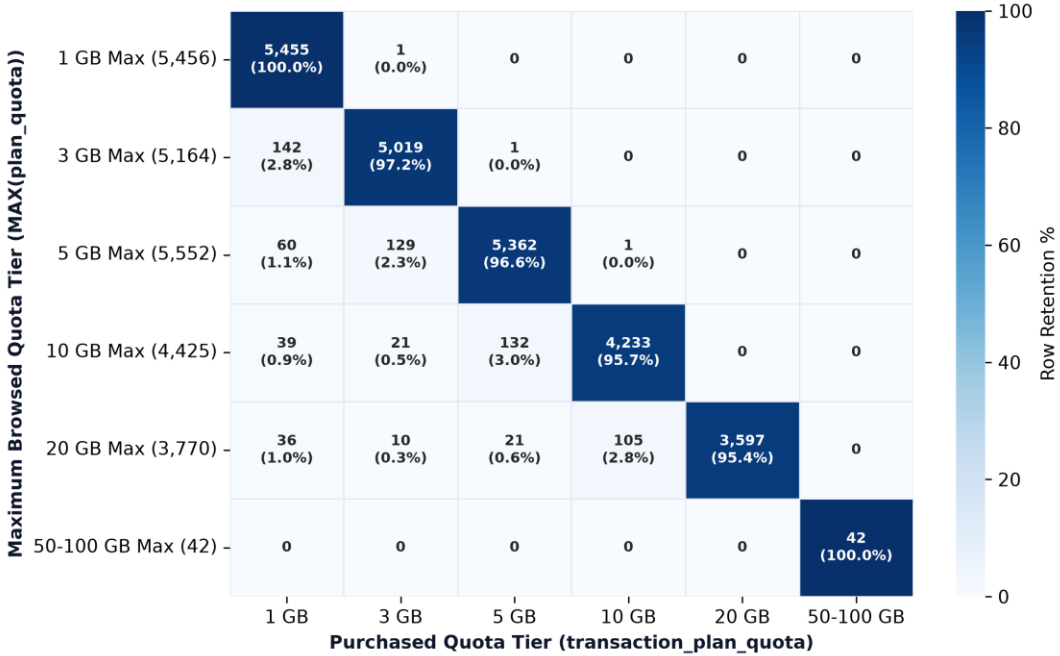


Behavioral Dimension	Segment / Tier Breakdown	Unique Sessions	Converting Buyers	Conversion Rate / Share	Product Engineering Action
Event Depth (Golden Window)	2–3 Events vs 7+ Events	74,546 vs 2,184	22,638 vs 129	30.37% CR (92.57% of buyers) vs 5.91%	Keep path to purchase strictly within 2–3 taps
Multi-Country Indecision (H8)	1 Country vs 2 vs 3+ Countries	91,117 / 14,603 / 10,275	21,970 / 1,770 / 687	24.11% -> 12.12% -> 6.69% (-72.3% drop)	Trigger Regional Bundle Card on 2nd country search
Destination Cross-Shopping	Viewed Alternative vs Single Dest.	3,078 vs 21,349 buyers	693 in EU <-> AT pair	12.59% cross-shop (#1 EU<->AT = 22.51%)	Compare EU ($16.65/5GB) vs AT ($9.79/5GB) inline
Checkout Retry Loops (S2)	iOS 1 Attempt vs 3+ Attempts	39,998 vs 736 at S2	14,987 vs 132 buyers	37.47% -> 17.92% CR (-52.18% drop)	Auto-open 1-Tap Payment Fallback on 2nd S2 tap

97.12% (23,723 / 24,426) Buy Their Maximum Browsed Quota (92.16% Single-Quota + 4.96% Multi-Quota Largest)

Module 4: Price Elasticity & Quota Exploration Matrix Across 24,426 Converting Sessions





1. SINGLE QUOTA VIEWED

92.16% (22,512 Buyers)

Viewed exactly 1 quota tier and bought it; zero mid-session quota switching

2. MULTI-QUOTA -> LARGEST

4.96% (1,211 Buyers)

Explored 2+ quota tiers and purchased the largest quota viewed (up-tier lock-in)

TOTAL DIAGONAL RETENTION

97.12% (23,723 Buyers)

22,512 single-quota + 1,211 multi-quota largest = 97.12% buy MAX(plan_quota)

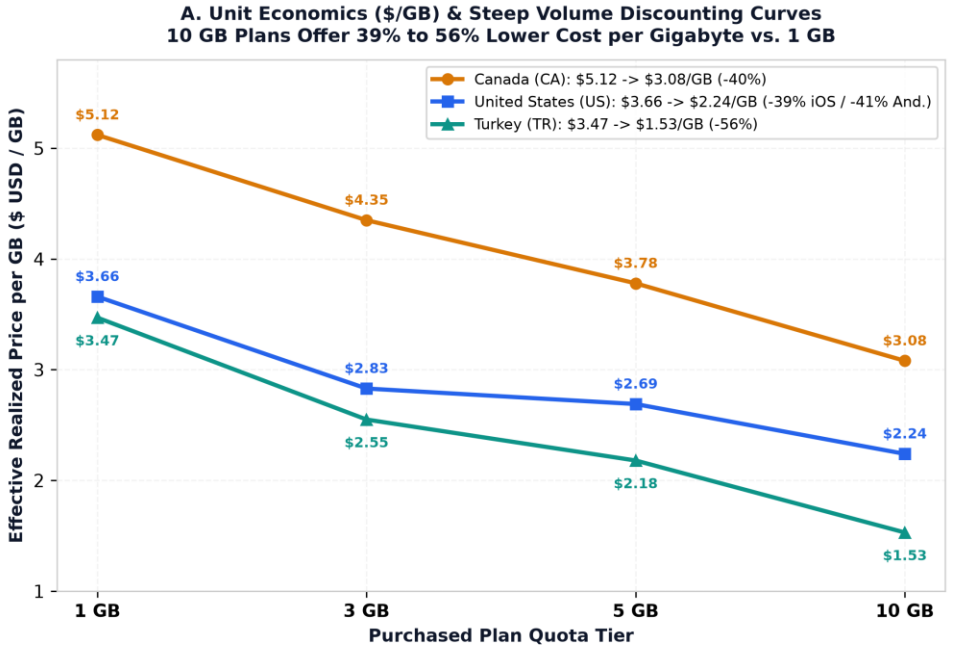
OFF-DIAGONAL DOWN-SELL

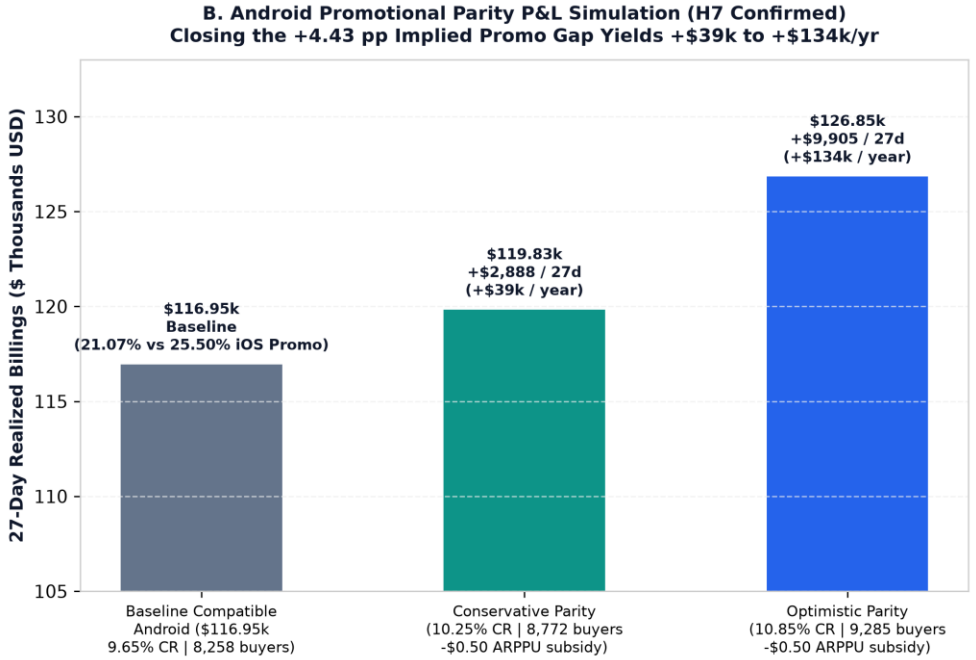
Only 2.88% (703 Buyers)

668 (2.73%) bought smallest viewed; 30 (0.12%) middle; 5 (0.02%) other -> H5 Rejected

10 GB Cuts $/GB by 39%–56%; Closing the +4.43 pp Promo Gap Yields +$39k–$134k/yr; Telemetry v2 Fixes 6 Schema Gaps

Module 4: Unit Economics ($/GB Elasticity), Implied Promo Asymmetry & Android Parity P&L (H7)





Telemetry v2 / Economic Pillar	Current v1 Schema / Baseline State	Proposed v2 Engineering / Growth Specification	Quantified Impact / Governance Value
1. Promo Asymmetry (H7 Confirmed)	25.50% iOS orders discounted ($1.29) vs 21.07% Android ($1.14)	Equalize Android promo deeplink parity (+0.60 to +1.20 pp CR lift)	+$2,888 to +$9,905 / 27d (+$39k to +$134k/yr)
2. Split fct_sessions & fct_orders	30,836 duplicate session IDs (33,147 surplus rows) in sessions	Enforce 1 row/unique_session_id in fct_sessions & 1 row/transaction_id in fct_orders	Eliminates -0.97 pp CR distortion (9.15% vs 10.12%)
3. Top-of-Funnel & Timestamp SDK	125,553 bounced sessions (51.98%) log 0 events; no timestamps	Add event_timestamp, app_open, compatibility_modal_shown, destination_search	Illuminates 70.78% Android Incompat bounce & path order
4. User Identity, Promo & Debounce	No user_id, no list_price_usd / promo_code_id, 250 tap loops	Add hashed user_id, list_price_usd, ad_cost_usd & 500ms client tap debounce	Enables multi-session LTV, exact ROAS & retry root cause